

SHORT QUESTIONS

In a diagram build interaction and provide declinations between AI and Data Science.

1. Differentiate between Machine Learning and Deep Learning
2. Differentiate between Traditional and Machine Learning Paradigm with two diagrams
3. Name three (03) algorithms for each of the following Learning paradigms:
 - a. Supervised Learning
 - b. Unsupervised Learning
 - c. Reinforcement Learning
4. Give the General equation for Multivariate Linear Regression with 04 Independents variables
5. Give the general Equation of the Logistic regression, and plot the sigmoid function
6. What is the purpose of splitting the Data Set into training and testing set
7. Differentiate between Bias and variance
8. Differentiate between Decision Tree and Random Forest
9. Give the formulae to calculate the accuracy and the recall in classification Problem

QUESTION

From the following Confusion Matrix

1. Identify TP, FP, TN, and FN
2. Give the formula and calculate:
 - a. False Positive Rate, TPR, TNR
 - b. Sensitivity, Precision
 - c. F1-Score, accuracy

	Predicted Positive	Predicted Negative
Actual Positive	85	12
Actual Negative	10	75

SECTION C - PRACTICALS :

QUESTION A

Considering the **Digits** dataset for optical character recognition (OCR) from SKLearn library.

Give the syntax to accomplish the following actions:

1. Import the logistic Regression Classifier
2. Split the Dataset in 75% for training
3. Train the Model
4. Evaluate the Score
5. Predict the corresponding values for the Test set
6. Generate the confusion Matrix corresponding

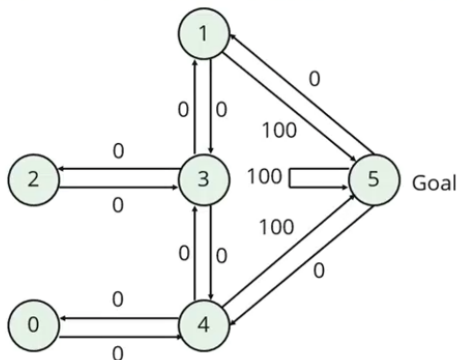
QUESTION B:

Import the **IRIS** dataset from **sklearn** library

1. Split the dataset with 15% allocated for the testing set.
2. Use logistic regression, Support Vector Machine and Decision Tree to generate the score for for each Model
3. Use the `cross_val_score` object with 6 slots, and Measure the different scores for logistic regression, SVM, Random forest, and decision Tree.
4. Write the script to give the average score for each model from the `cross_val_score`
5. Plot the confusion Matrix for the Random forest model

Question

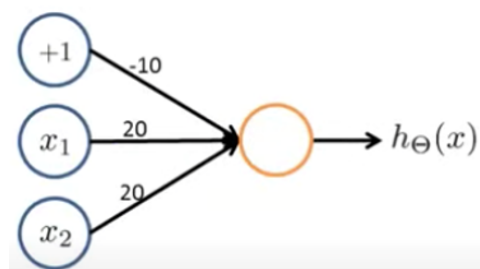
Given the following State diagram, in a reinforcement Learning based architecture



1. Give the corresponding R Matrix to this state diagram
2. Give the $Q(\text{State}, \text{Action})$ general equation
3. Calculate the $Q(4,5)$ considering $\text{Gamma} = 0.4$

Question :

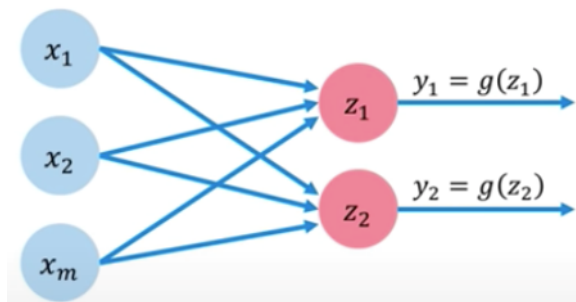
Given the figure by aside presenting a perceptron, with a sigmoid activation function,



- 1) Write the $h_{\theta}(x)$ equation
- 2) Build the truth table assuming x_1 and x_2 are binary inputs and deduce the function implemented through $h_{\theta}(x)$.

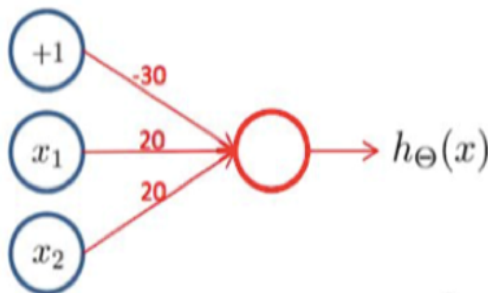
Question

Given the following ANN diagram



4. Anotate the various weights
5. Give the equation for z_1
6. Give the equation for y_1 given that the activation function $g()$ is Sigmoid
7. Generalize the z_i
8. State the Chain Rule principle

Question



Given the figure by aside presenting a perceptron, with a sigmoid activation function,

3) Write the $h_{\theta}(x)$ equation

4) Build the truth table assuming x_1 and x_2 are binary inputs and deduce the fuction implemented through $h_{\theta}(x)$.

Question

Given the following topics in deep Learning, apply a use case implementation for the the following topics:

- **Topic 1:** CNN models with CIFAR
- **Topic 2:** Transfer Learning for Object recognition
- **Topic 3:** Real Time objects Counter with Yolo
- **Topic 4:** Generatives Models with RAGs.
- **Topic 5:** Build a Model to regonise 5 objets using YOLO
- **Topic 6:** Build a Model to regonise 5 objets using TEACHABLE MACHINE/MediaPipe
- **Topic 7:** Build a Model For SPEECH TO TEXT translation
- **Topic 8:** Build a Model For TEXT TO SPEECH translation
- **Topic 9:** Build a Model For SENTIMENT ANALYSIS
- **Topic 10:** Build a PIPELINE For SPEECH TO TEXT -> SENTIMENT ANALYSIS -> TEXT SPEECH
- **Topic 10:** Build a Model based Q-Learning for Labyrinth Navigation

Provide the various steps for data Preparation

Build the Architecture' PIPELINE

Provide the various steps for Building the model

Sketch the Network's chosen architecture

Provide Key Source (Or Pseudo Code) for Training and Evaluation

State the Key evaluation Metrics

Section B : Practicals [30 Marks]

1. Run an example of application for the choosen topic in the last question from
 - **Topic 1:** CNN models with CIFAR
 - **Topic 2:** Transfer Learning
 - **Topic 3:** Real Time objects Counter with Yolo
 - **Topic 4:** Generatives Models with RAGs.
2. NB: Set some callBack(s) checkings to control the training phase where applicable
3. Provide the correct evaluation metrics, and plot it LOSS, and ACCURACY of the Training and Validation for the training phase where applicable.
4. Evaluate the score or provide the demo of the model on the testing set.

SECTION A: MCQ IN AI

Artificial Intelligence (AI)

1. What is the primary goal of Artificial Intelligence?
 - a) To replace human intelligence
 - b) To create machines that can perform tasks requiring human intelligence
 - c) To develop only theoretical models
 - d) To focus solely on robotics
2. Which of the following is NOT a type of AI?
 - a) Narrow AI
 - b) General AI
 - c) Super AI
 - d) Reactive AI
3. What is the Turing Test used for?
 - a) To measure the speed of a computer
 - b) To determine if a machine can exhibit intelligent behavior
 - c) To evaluate the hardware performance of a system
 - d) To test the security of an AI system
4. Which of the following is an example of a rule-based AI system?
 - a) Neural networks
 - b) Expert systems
 - c) Reinforcement learning
 - d) Deep learning
5. What is the main limitation of rule-based AI systems?
 - a) They require large amounts of data
 - b) They cannot handle uncertainty or ambiguity
 - c) They are too slow
 - d) They are only applicable to robotics

Machine Learning (ML)

6. What is the primary purpose of Machine Learning?
 - a) To program explicit rules for decision-making
 - b) To enable systems to learn from data and improve over time
 - c) To replace human decision-making entirely
 - d) To focus only on data storage
7. Which of the following is a supervised learning algorithm?

- a) K-Means Clustering
- b) Decision Trees
- c) Principal Component Analysis (PCA)
- d) Apriori Algorithm

8. What is the role of a cost function in Machine Learning?

- a) To evaluate the performance of a model
- b) To preprocess the data
- c) To visualize the data
- d) To store the dataset

9. Which of the following is an example of unsupervised learning?

- a) Linear Regression
- b) Logistic Regression
- c) K-Means Clustering
- d) Support Vector Machines (SVM)

10. What is overfitting in Machine Learning?

- a) When a model performs well on training data but poorly on unseen data
- b) When a model is too simple to capture patterns in the data
- c) When a model fails to converge during training
- d) When a model is trained on insufficient data

1. What is the primary goal of supervised learning?

- a) To find patterns in unlabeled data
- b) To predict outcomes based on labeled data
- c) To reduce the dimensionality of data
- d) To cluster data into groups

2. Which of the following is an example of a classification algorithm?

- a) Linear Regression
- b) K-Means Clustering
- c) Decision Trees
- d) Principal Component Analysis (PCA)

3. What is overfitting in machine learning?

- a) When a model performs well on training data but poorly on unseen data
- b) When a model performs poorly on both training and test data
- c) When a model is too simple to capture patterns in the data
- d) When a model has too few features

4. Which evaluation metric is used for regression problems?

- a) Accuracy
- b) F1 Score
- c) Mean Squared Error (MSE)
- d) Precision

5. What is the purpose of the activation function in a neural network?

- a) To reduce the dimensionality of the data
- b) To introduce non-linearity into the model
- c) To normalize the input data
- d) To optimize the learning rate

6. Which of the following is a unsupervised learning algorithm?

- a) Logistic Regression
- b) K-Means Clustering
- c) Support Vector Machines (SVM)
- d) Random Forest

7. What is the role of the learning rate in gradient descent?

- a) It determines the number of iterations required for convergence
- b) It controls the step size during parameter updates
- c) It defines the number of layers in a neural network
- d) It measures the model's accuracy

8. Which of the following is NOT a type of cross-validation technique?

- a) K-Fold Cross-Validation
- b) Leave-One-Out Cross-Validation
- c) Random Subsampling
- d) Principal Component Analysis (PCA)

9. What is the main advantage of using ensemble methods?

- a) They reduce the need for labeled data
- b) They combine multiple models to improve performance
- c) They are faster to train than single models
- d) They require less computational resources

10. Which algorithm is used for dimensionality reduction?

- a) Logistic Regression
- b) K-Nearest Neighbors (KNN)
- c) Principal Component Analysis (PCA)
- d) Decision Trees

11. What is the purpose of regularization in machine learning?

- a) To increase model complexity
- b) To reduce overfitting by penalizing large coefficients
- c) To speed up the training process
- d) To improve model interpretability

12. Which of the following is a hyperparameter in a neural network?

- a) Weights
- b) Bias
- c) Learning rate
- d) Activation function

13. What is the output of a logistic regression model?

- a) A continuous value
- b) A probability value between 0 and 1
- c) A binary value (0 or 1)
- d) A categorical label

14. Which of the following is true about the bias-variance tradeoff?

- a) High bias leads to underfitting, and high variance leads to overfitting
- b) High bias leads to overfitting, and high variance leads to underfitting
- c) Both high bias and high variance lead to underfitting
- d) Both high bias and high variance lead to overfitting

15. What is the purpose of the confusion matrix?

- a) To evaluate the performance of a regression model
- b) To visualize the performance of a classification model
- c) To reduce the dimensionality of the data
- d) To optimize hyperparameters

16. Which of the following is a loss function used in linear regression?

- a) Cross-Entropy Loss
- b) Hinge Loss
- c) Mean Squared Error (MSE)
- d) Log Loss

17. What is the main difference between bagging and boosting?

- a) Bagging trains models in parallel, while boosting trains models sequentially
- b) Bagging reduces bias, while boosting reduces variance
- c) Bagging is used for regression, while boosting is used for classification
- d) Bagging uses weighted data, while boosting uses equal weights

18. Which of the following is a generative model?
- a) Logistic Regression
 - b) Naive Bayes
 - c) Support Vector Machines (SVM)
 - d) Decision Trees
19. What is the purpose of the dropout layer in a neural network?
- a) To reduce overfitting by randomly dropping neurons during training
 - b) To increase the number of layers in the network
 - c) To normalize the input data
 - d) To speed up the training process
20. Which of the following is NOT a clustering algorithm?
- a) K-Means
 - b) DBSCAN
 - c) Hierarchical Clustering
 - d) Linear Regression

Deep Learning

11. What is the primary advantage of deep learning over traditional ML?
- a) It requires less computational power
 - b) It can automatically extract features from raw data
 - c) It does not require labeled data
 - d) It is easier to interpret
- Answer: b) It can automatically extract features from raw data
12. Which of the following is a popular deep learning framework?
- a) Scikit-learn
 - b) TensorFlow
 - c) Weka
 - d) RapidMiner
13. What is a neural network?
- a) A set of rules for decision-making
 - b) A computational model inspired by the human brain
 - c) A database management system
 - d) A type of clustering algorithm
14. What is the purpose of an activation function in a neural network?
- a) To normalize the input data
 - b) To introduce non-linearity into the model
 - c) To reduce the dimensionality of the data
 - d) To store the weights of the network

15. Which of the following is NOT a type of neural network?

- a) Convolutional Neural Network (CNN)
- b) Recurrent Neural Network (RNN)
- c) Decision Tree
- d) Generative Adversarial Network (GAN)

Applications and Ethics

16. Which of the following is an application of Natural Language Processing (NLP)?

- a) Image recognition
- b) Sentiment analysis
- c) Fraud detection
- d) Clustering

17. What is the main ethical concern regarding AI and ML?

- a) High computational cost
- b) Bias in algorithms
- c) Lack of data availability
- d) Complexity of models

18. Which of the following is an example of reinforcement learning?

- a) Training a robot to navigate a maze
- b) Classifying emails as spam or not spam
- c) Grouping customers based on purchasing behavior
- d) Predicting house prices

19. What is the purpose of cross-validation in ML?

- a) To reduce the size of the dataset
- b) To evaluate the model's performance on unseen data
- c) To preprocess the data
- d) To visualize the data

20. Which of the following is a key challenge in deploying AI systems?

- a) Lack of algorithms
- b) Data privacy and security
- c) High cost of hardware
- d) All of the above